

CS215 Assignment-1

Malay Kedia, Aakash Gupta, Yash Singh

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1. Finding optimal bandwidth

Task 1

Part (a)

Solution:

We are asked to prove the following:

$$\int \hat{f}(x)^2 dx = \frac{1}{n^2 h} \sum_{j=1}^m v_j^2$$

where $\hat{f}(x)$ is the histogram estimator, v_j is the number of points in the j -th bin, and h is the bin width.

Proof

The histogram estimator $\hat{f}(x)$ is defined as:

$$\hat{f}(x) = \sum_{j=1}^m \frac{p_j}{h} \mathbf{1}(x \in B_j)$$

also,

$$p_j = \frac{v_j}{n}$$

so the expression becomes,

$$\hat{f}(x) = \sum_{j=1}^m \frac{v_j}{nh} \mathbf{1}(x \in B_j)$$

where: - v_j is the number of data points in the j -th bin. - $\mathbf{1}(x \in B_j)$ is the indicator function that equals 1 if x belongs to bin B_j and 0 otherwise.

Next, we need to compute $\hat{f}(x)^2$:

$$\hat{f}(x)^2 = \left(\sum_{j=1}^m \frac{v_j}{nh} \mathbf{1}(x \in B_j) \right)^2$$

Since the indicator functions $\mathbf{1}(x \in B_j)$ are mutually exclusive (a point x can only belong to one bin at a time), we can ignore cross-terms in the expansion of the square, leading to,

$$\hat{f}(x)^2 = \sum_{j=1}^m \left(\frac{v_j}{nh} \right)^2 \mathbf{1}(x \in B_j)$$

Now, we integrate $\hat{f}(x)^2$ over the entire domain of x :

$$\int \hat{f}(x)^2 dx = \int \sum_{j=1}^m \left(\frac{v_j}{nh} \right)^2 \mathbf{1}(x \in B_j) dx$$

Since the bins are disjoint, the integral over each bin B_j simplifies to the width of the bin, which is h . Therefore, the integral becomes:

$$\int \hat{f}(x)^2 dx = \sum_{j=1}^m \left(\frac{v_j}{nh} \right)^2 h$$

Simplifying further:

$$\int \hat{f}(x)^2 dx = \frac{1}{n^2 h} \sum_{j=1}^m v_j^2$$

Thus, we have proven that:

$$\int \hat{f}(x)^2 dx = \frac{1}{n^2 h} \sum_{j=1}^m v_j^2$$

Part (b)

Solution:

We need to prove the following expression:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m (v_j^2 - v_j)$$

where $\hat{f}_{(-i)}$ is the histogram estimator after removing the i -th observation, and v_j is the number of points in the j -th bin.

Proof

The histogram estimator $\hat{f}(x)$ for the complete dataset is given by:

$$\hat{f}(x) = \sum_{j=1}^m \frac{v_j}{nh} \mathbf{1}(x \in B_j)$$

where v_j is the number of points in the j -th bin.

When we remove the i -th observation, the number of points in the bin containing X_i is reduced by 1. Therefore, the histogram estimator after removing the i -th point which maybe in any of the bin as indicated by introduction of predicate for the same, $\hat{f}_{(-i)}(x)$, is:

$$\hat{f}_{(-i)}(x) = \sum_{j=1}^m \frac{v_j - \mathbf{1}(X_i \in B_j)}{(n-1)h} \mathbf{1}(x \in B_j)$$

Now, sum the values of $\hat{f}_{(-i)}(X_i)$ for all i :

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \sum_{i=1}^n \sum_{j=1}^m \frac{v_j - \mathbf{1}(X_i \in B_j)}{(n-1)h} \mathbf{1}(X_i \in B_j)$$

For each bin j , the indicator function $\mathbf{1}(X_i \in B_j)$ equals 1 if X_i is in bin j , and 0 otherwise. Therefore, for each bin j , the sum over i simplifies to:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m \sum_{i \in B_j} (v_j - 1)$$

For each bin j , there are v_j points in total. Thus, the sum over $i \in B_j$ yields v_j terms, each equal to $v_j - 1$. Hence, the inner sum simplifies to:

$$\sum_{i \in B_j} (v_j - 1) = v_j(v_j - 1)$$

Substituting this into the previous expression, we get:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m v_j(v_j - 1)$$

Expanding the product $v_j(v_j - 1)$, we get:

$$v_j(v_j - 1) = v_j^2 - v_j$$

Thus, the final expression becomes:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m (v_j^2 - v_j)$$

Task 2

Part(a)

Solution:

The probability distributions p_j for all the bins as calculated in the code as $\frac{v_j}{n}$ are as follows:

Bin 1 [0.382 - 0.7238]: Probability = 0.2059

Bin 2 [0.7238 - 1.0655999999999999]: Probability = 0.4882

Bin 3 [1.0655999999999999 - 1.4074]: Probability = 0.0471

Bin 4 [1.4074 - 1.7492]: Probability = 0.0412

Bin 5 [1.7492 - 2.091]: Probability = 0.1353

Bin 6 [2.091 - 2.4328]: Probability = 0.0588

Bin 7 [2.4328 - 2.7746]: Probability = 0.0059

Bin 8 [2.7746 - 3.1164]: Probability = 0.0000

Bin 9 [3.1164 - 3.4582]: Probability = 0.0118

Bin 10 [3.4582 - 3.8]: Probability = 0.0059

Part(b)

Solution: Based on probability distributions of the bins, maximum data (almost 95%) falls within bin 1. This is due to very large bandwidth of the history and hence the histogram is **oversmooth**.

Part(d)

Solution: Optimal bandwidth is obtained where cross validation score is minimum. From the curve, this is achieved when bandwidth is **around 0.08**.

1. Higher Order Regression

Answer 1

Solution: In a simple linear regression model, the equation of the least squares regression line is:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where symbols have their usual meanings.

The least squares estimates for $\hat{\beta}_1$ and $\hat{\beta}_0$ are given by:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i Y_i - \bar{x} \sum_{i=1}^n Y_i}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x}$ and $\bar{Y}$ are the means of x and Y , respectively.

We want to show that the point $(\bar{x}, \bar{Y})$ lies on the regression line. This means that when $x = \bar{x}$, the predicted value $\hat{Y}$ should be equal to $\bar{Y}$.

Substitute $x = \bar{x}$ into the equation of the regression line:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Now, substitute the expression for $\hat{\beta}_0$:

$$\hat{Y} = (\bar{Y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x}$$

Simplifying this:

$$\begin{aligned}\hat{Y} &= \bar{Y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x} \\ \hat{Y} &= \bar{Y}\end{aligned}$$

Thus, when $x = \bar{x}$, the predicted value $\hat{Y}$ equals $\bar{Y}$.

Hence, the point $(\bar{x}, \bar{Y})$ always lies on the least squares regression line, as the predicted value $\hat{Y}$ equals the mean value $\bar{Y}$ when $x = \bar{x}$.

Answer 2

Solution: The least squares estimates for β_0 and β_1 can be derived by minimizing the sum of squared residuals:

$$\text{Residual sum of squares (RSS)} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

The least squares estimates are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i Y_i - \bar{x} \sum_{i=1}^n Y_i}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x}$ and $\bar{Y}$ are the means of x and Y , respectively.

Now, suppose we use $z = x - \bar{x}$ as the regressor variable. The new model becomes:

$$Y = \beta_0^* + \beta_1^* z + \epsilon$$

Since $z = x - \bar{x}$, this new model can be rewritten as:

$$Y = \beta_0^* + \beta_1^*(x - \bar{x}) + \epsilon$$

Expanding this:

$$Y = \beta_0^* + \beta_1^* x - \beta_1^* \bar{x} + \epsilon$$

We can compare this to the original model $Y = \beta_0 + \beta_1 x + \epsilon$. Clearly, the coefficient β_1^* is the same as β_1 , i.e.,

$$\beta_1^* = \beta_1$$

Now, to match the original model, we can identify:

$$\beta_0 = \beta_0^* - \beta_1 \bar{x}$$

Thus, the intercept in the transformed model, β_0^* , relates to the intercept in the original model, β_0 , as:

$$\beta_0^* = \beta_0 + \beta_1 \bar{x}$$

Hence the following inferences can be drawn regarding relations between the least square estimates of the two models:

- The slope β_1^* is the same as β_1 : $\beta_1^* = \beta_1$.
- The intercept β_0^* is shifted by $\beta_1 \bar{x}$: $\beta_0^* = \beta_0 + \beta_1 \bar{x}$.

Differences between the two models are as follows:

- The new model, by centering the x values (i.e., subtracting $\bar{x}$ from each x_i), makes the interpretation of the intercept β_0^* simpler as β_0^* represents the value of Y when $x = \bar{x}$, i.e., when $z = 0$.
- In the original model, β_0 represents the expected value of Y when $x = 0$, which might be less meaningful depending on the context (if 0 is outside the range of observed x values).

Thus, the transformation to $z = x - \bar{x}$ simplifies the interpretation of the intercept but leaves the slope unchanged.